AWS Payment System Architecture

Next-Generation Multi-Channel Processing Engine

Document Category: Infrastructure Architecture Design

Target Channels: Business-to-Business (B2B) & Business-to-Customer (B2C)

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Classification: Restricted Confidential

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1. Executive Summary & Core Channels

This document details the target cloud architecture deployed on Amazon Web Services (AWS) to govern high-volume payment transactions across dual portfolios: Business-to-Business (B2B) bulk financing settlement and Business-to-Customer (B2C) real-time micro-payments. The framework leverages highly segregated entry barriers, modular compute tiers, and distinct database targets optimized for throughput and rigorous compliance metrics.

1.1 Business-to-Customer (B2C) Transaction Flow

The consumer-facing path facilitates microtransactions originating from web and native smartphone portals. High availability and instantaneous validations are achieved via AWS edge security features. Traffic is ingestion-throttled through Amazon API Gateway endpoints before parsing authentication parameters inside fully localized AWS Cognito User Pools.

Because consumer behavioral payloads demand rapid execution loops, operations utilize an decoupled architecture. Ephemeral messages settle directly into streaming microservices, mitigating downstream pressure on transactional data lakes during flash event spikes.

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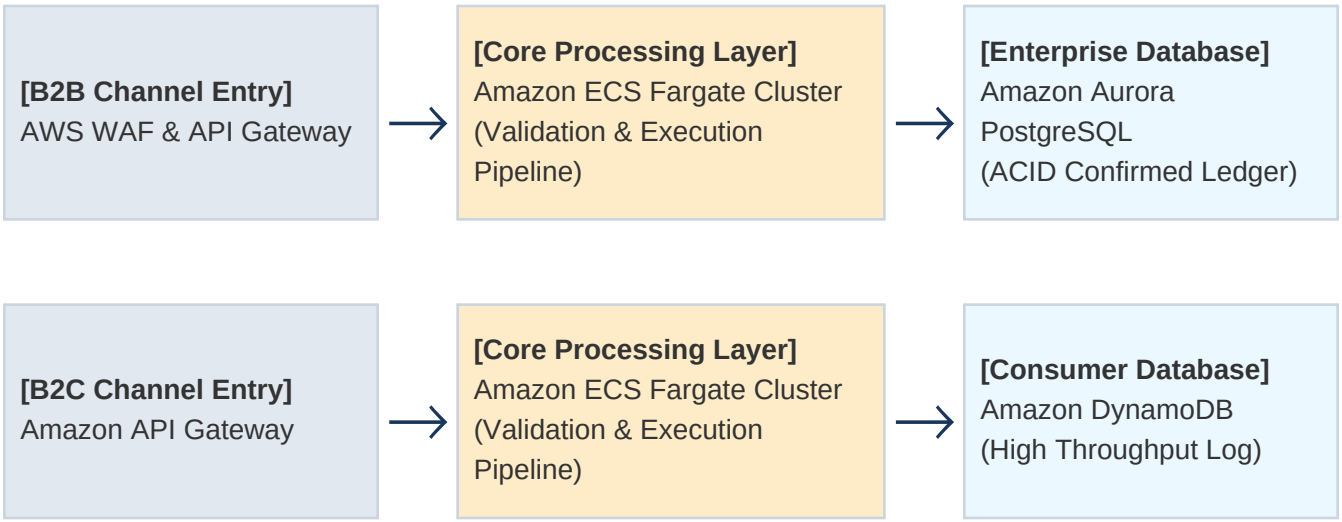
2. Business-to-Business (B2B) Processing & Ingestion

The enterprise infrastructure framework prioritizes high reliability, large payload compliance, and extensive validation pipelines over raw transaction speed. Corporate ledger adjustments often span millions of dollars per transaction line item, necessitating strict isolation controls and dedicated virtual private networks.

2.1 B2B Transaction Ingestion Pipeline

Enterprise transactions interface via designated REST and gRPC API points safeguarded by AWS WAF rulesets and private corporate VPC connections. Once authenticated against specialized AWS IAM roles and mutual TLS validation layers, incoming payment batches enter a strict processing layer within an isolated Amazon ECS environment.

Figure 2.2: Infrastructure Ingestion Diagrammatic Mapping



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3. Database Models & Core Service Matrix

Storage selection reflects the operational profiles of each distinct transaction engine. High-frequency consumer workloads demand horizontally scalable distributed ledgers, while corporate batch files dictate ACID compliance and complex relational integrity models.

Component Name	AWS Service Target	Primary Functionality	Performance Driver
B2C Entry Point	Amazon API Gateway	Throttling, Edge Security, Ingestion	Low Latency Real-time API Hooks
B2B Core Compute	Amazon ECS (Fargate)	Corporate Payload Processing	High Capacity & Isolation
B2C Ledger Lake	Amazon DynamoDB	High-speed NoSQL Key-Value Store	Single-Digit Millisecond Scale Access
B2B Ledger Lake	Amazon Aurora PostgreSQL	Relational Corporate Double-Entry	Strict ACID Compliance & Complex Joins
Identity Control	AWS Cognito / IAM	Role-Based Entitlements & Tokens	Unified Access & Encryption

3.1 Summary of Controls

The alignment between computing layers and isolated multi-region databases provides security redundancy across both enterprise and retail portals. Future infrastructure updates will encompass real-time fraud mitigation powered by streaming analytical pipelines seamlessly attached to data storage endpoints.

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